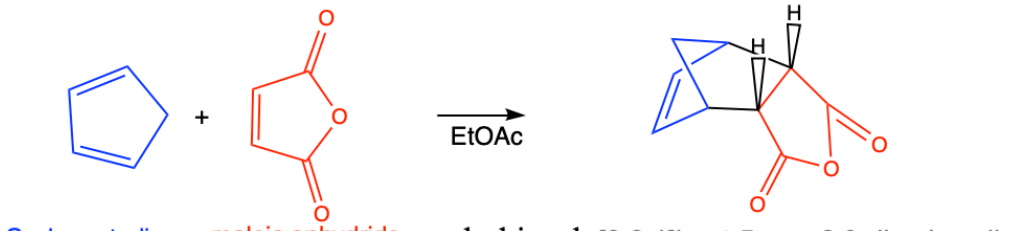
Diels Alder Reaction Cyclopentadiene with Maleic anhydride

In Diels-Alder reaction a conjugated diene (a molecule with two adjacent double bonds) reacts with an alkene (called a dienophile) to form a six-membered ring, creating two new carbon-carbon bonds through a concerted mechanism (mechanism in which all bond breaking and forming occurs simultaneously in a cyclic transition state). For the Diels-Alder to occur the diene must be in the s-cis conformation, the pi-bonds of the diene should be adjacent and in the same plane.

The Diels-Alder reaction between cyclopentadiene and maleic anhydride occurs as a concerted, single-step process where the diene (cyclopentadiene) attacks the dienophile (maleic anhydride) through a cyclic transition state forming endo-bicyclo[2.2.1]hept-5-ene-2,3-dicarboxylic anhydride (also known as 5-Norbornene-2,3-dicarboxylic anhydride). Major product is an endo adduct where the anhydride group is oriented closer to the other substituents on the cyclopentadiene ring, due to secondary orbital interactions between the diene and the electron-withdrawing carbonyl groups of the anhydride.
This reaction is particularly fast due to the electron-rich diene and electron-deficient dienophile nature of the reactants.



Cyclopentadiene maleic anhydride endo-bicyclo[2.2.1]hept-5-ene-2,3-dicarboxylic anhydride

M_w: 66.10 g/mol	M_w: 98.06 g/mol	M_w: 164.16 g/mol

In the experiment ethyl acetate and hexane will be used as solvents. Ethyl acetate, a polar a protic solvent, helps solubilize the reactants and promotes their interaction during the reaction. Additionally, its relatively low boiling point (approximately 77 °C) allows for easy removal during the workup process.
Hexane a nonpolar solvent is used to wash the reaction mixture, aids in the separation and purification of the desired product.

Procedure:

- Take 3 mL of hexanes in a reaction tube. Cool it in an ice bath.
- Weigh 0.20 g Maleic anhydride in a reaction tube or vial.((Be VERY Careful with Maleic Anhydride-keep below your waist)
- Add 1 mL Ethyl Acetate to dissolve maleic anhydride
- Add 0.2 mL of cyclopentadiene to the reaction with stirring in the hood.
- Stir for 10 minutes.
- Add 1 mL of cold hexane dropwise with swirling.
- Product should crystallize.
- Cool for 10 min.
- Decant the excess solvent. Remove the solvent from the crystals with a Pasteur pipette.
- Scrape the product onto a piece of filter paper, allow the crystals to air-dry, determine their weight, and calculate the yield of the product.
- Determine the melting point of the product and turn in any material not used in the experiment.

For the IR spectrum and the H1 NMR spectrum of the anhydride adduct, refer to Figure 48.8 and Figure 48.9 respectively, Chapter 11, of the textbook, Williamson, K.L., Masters, K.M. Macroscale and Microscale Organic Experiments, 7th Edition, Cengage.

Cleaning up

Place the crystallization solvent mixture in the organic solvents waste container. Allow the organic material to evaporate from the drying agent/ filter paper (in the hood); then place it in the nonhazardous solid waste container.

Ref: eTextbook: Macroscale and Microscale Organic Experiments Cengage Learning Book Updated as of - January 6, 2025, page 626.
(https://ebooks.cengage.com/reader/d7b768e6-07f5-43e2-9a65-d98e97859942/content-bd_ch_48_opener?code=3FXF73TyTOgM2WbXlLhvyOlBkUWunDpXMDzaJAIZiTQ&state=005a6b95-3823-45b5-9d1a-810359527c10. Accessed: 09/13/2026)